Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years building customer integrations, **APIs**, and AI **document** workflows in **TypeScript** and Python, working directly with clients to scope requirements, debug across systems, and ship production software end to end.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Built **ETL-style integrations** connecting FastAPI services, Supabase, SQL Server, and cloud storage for a deployed AI platform, using incremental loads and reconciliation checks to run 5K+ monthly jobs at 99%+ reliability.
- Designed multi-tenant storage on Azure SQL with **row-level security** for 500+ users, cutting dashboard latency from 1.2s to 350ms, and restructured async LLM orchestration processing 1K+ jobs at 3x throughput.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built an AI **document analysis** pipeline with secure PDF, DOCX, and ZIP ingestion, vision OCR, and LLM evaluation across GPT-4o, Claude, and Qwen-VL, processing 1K+ files per batch for 150+ instructors.
- Shipped React and Flask REST features end to end to production on AWS, handling 10K+ daily requests with stable peak-load responses and **debugging** OCR timeouts with fallback routing, cutting turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Troubleshot large-scale **API integrations** with Microsoft and GitHub stakeholders, tracing failures through logs and tests, then adding pagination, field filtering, and caching to cut response times 30%.
- Took requirements through deployment on cross-platform data-sharing microservices, building **reusable** Singleton and Factory modules and 20+ automated tests that cut technical debt 20% and manual data entry 35%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** and Node.js microservices platform with JWT **authentication**, role-based access, and REST APIs for customer records, billing, and leads, syncing 2M+ records across locations in real time.
- **Debugged production issues** with Sentry, Azure Monitor, and Kibana logs and shipped fixes through Jenkins CI/CD with automated tests and staged approval gates, cutting deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Worked directly with the client to map 12+ integration points and **presented the end-to-end architecture** for approval, then built Flask booking, pricing, and coupon APIs with sub-200ms responses.
- Integrated card, wallet, and PayPal payment providers through **idempotent webhooks**, automating refunds and disputes to cut manual handling about 70%, and added wallet-signature authentication for passwordless login.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, Bash
Integrations & APIs: REST APIs, Webhooks, JWT, Role-Based Access, Microservices, Node.js, Express.js, FastAPI, Flask
AI & Documents: LLM Systems, OCR, Prompt Engineering, Embeddings, FAISS, Semantic Search, Document Ingestion
Data & Cloud: PostgreSQL, SQL Server, Azure SQL, Supabase, ETL Pipelines, AWS, Azure, Docker
Delivery & Debugging: React, Jest, PyTest, Integration Testing, Sentry, Kibana, Log Analysis, CI/CD, Git, Customer Communication

Projects

Document Management Connector Framework | *TypeScript, Node.js, Microsoft Graph, OAuth 2.0, Webhooks, PostgreSQL*

- Designed reusable **TypeScript** connectors that sync folders and documents from SharePoint and similar systems via OAuth, using delta queries, webhook change notifications, retries, and **permission mirroring** for secure access.

Contract Review Workflow Agent | *TypeScript, Python, LLM APIs, OCR, Embeddings, PostgreSQL*

- Designed an **LLM workflow** that extracts key clauses from large contract sets into a review table, citing source pages for every answer, with an eval set comparing outputs against **lawyer-labeled** examples.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, PostgreSQL, REST APIs*

- Built an AI document platform with Clerk authentication, transformer embeddings, and FAISS **semantic search** over 50K+ documents, enabling contextual Q&A and summarization while cutting document review time 60%.